

SEA QUANTATHON 2026 · DEEP-TECH QUANTUM COMPUTING

SYNDRA

Quantum-Learning Decoders for Stabilizer Codes
— toward Fault-Tolerant Quantum Computers

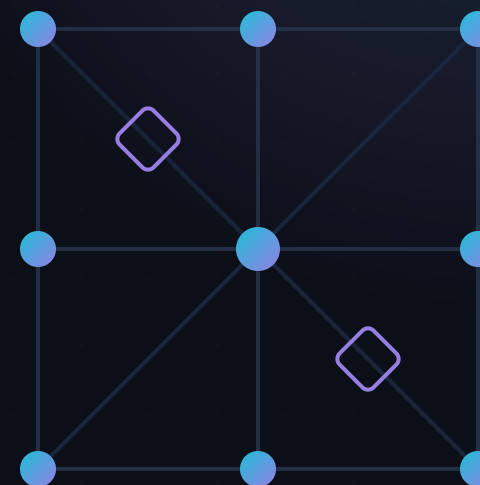
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QCNN decoding

VQE state prep

QGM generalization

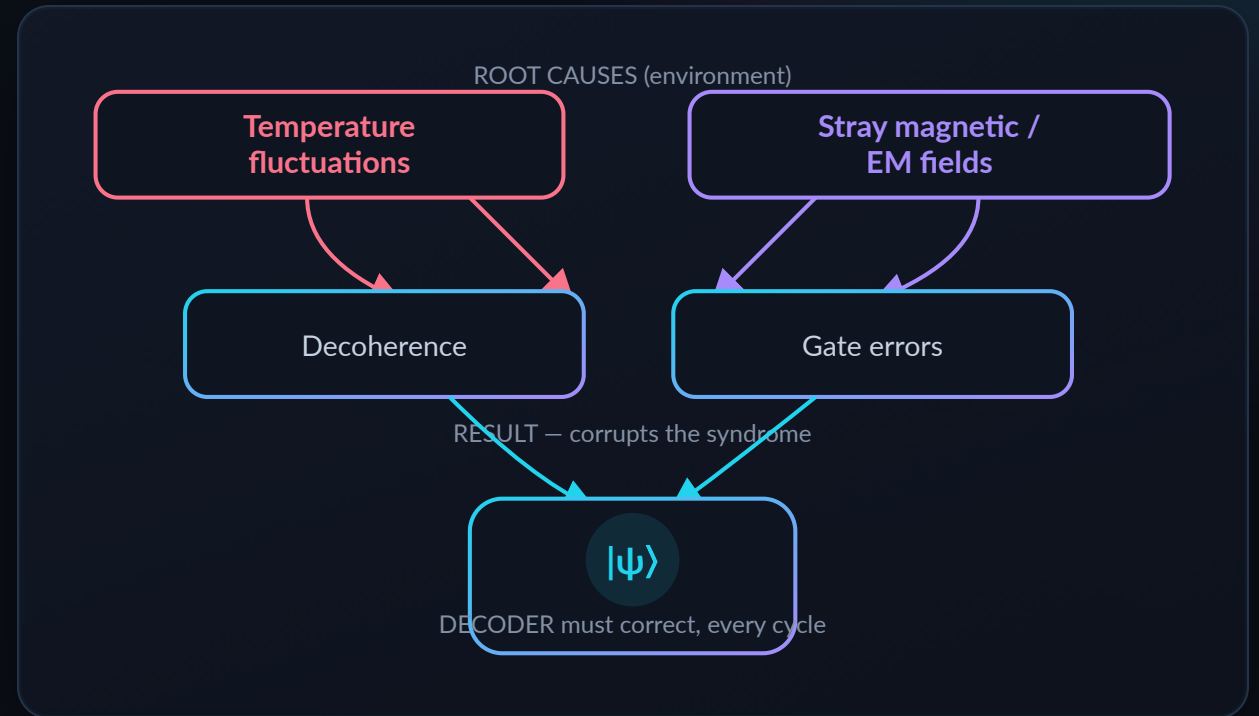
\$16.3B market by 2030



Noise corrupts qubits — decoding is how we fix it

Environmental sources — temperature fluctuations, stray magnetic/EM fields — **cause** decoherence and gate errors, which corrupt qubits.

- ▶ Fault-tolerant QC tolerates these errors **only if** a decoder reads the syndrome and corrects fast enough, every cycle.
- ▶ Classical decoders (MWPM) don't scale well and add quantum–classical round-trip latency that eats coherence time.
- ▶ **Decoding is the bottleneck** standing between today's noisy devices and useful fault-tolerant quantum computers.

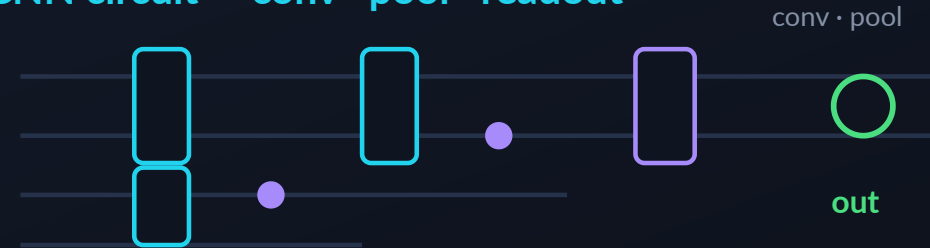


Root causes → decoherence/gate errors → a decoder must catch it before it propagates.

Quantum-learning decoders that act on the state directly

- ▶ **Quantum-native:** operate on quantum states directly — no quantum–classical round trip, adapts to real hardware noise.
- ▶ **Scalable ansatz:** QCNN uses $O(\log N)$ parameters, barren-plateau-free (Cong et al.) — trains where generic QML fails.
- ▶ **Staged rollout:** QCNN (decoding, today) → VQE (adaptive state prep) → QGM (cross-hardware generalization).
- ▶ **Hardware-agnostic:** retrains to any noise profile — superconducting, photonic, trapped-ion.

QCNN circuit — conv · pool · readout



Why this wins: a 12–16 qubit QCNN needs **~42–4,000** trainable parameters versus millions for classical NN decoders — cheaper to train, run, and adapt per hardware target.

Syndrome
detector clicks



QCNN decoder
learned, quantum-native

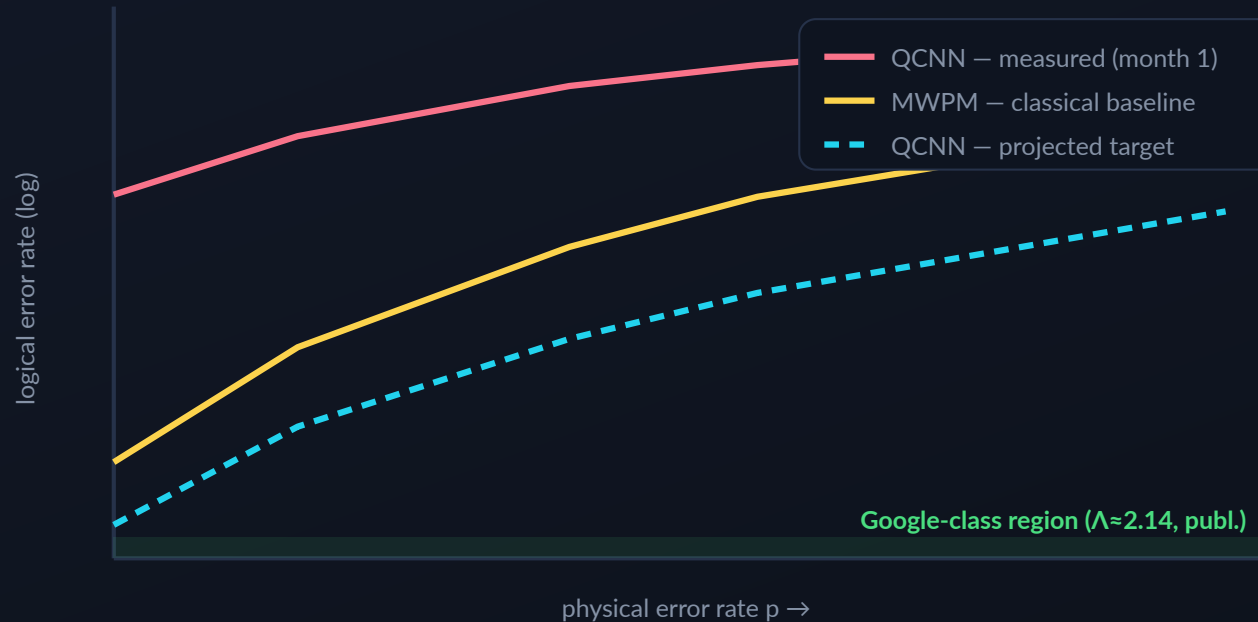


Logical flip?
correct / wrong

RESULTS

Measured baseline & target trajectory

Logical error rate p_L vs physical error rate p ($d=3$, 20k shots)



Solid = measured. **Dashed** = projected target (not yet measured).

Suppression factor Λ (>1 = scales)

- ▶ **MWPM 1.34** measured
- ▶ **QCNN today 0.68** measured, M1
- ▶ **Our target $\rightarrow 2.2$** projected
- ▶ **Google 2.14** published, 2024

HONEST BASELINE Full pipeline & metrics built first; gap quantified transparently. Target trajectory backed by sourced levers (geometry encoding, 10 $\times$ sim speedup).

Fault-tolerant QC is entering its infrastructure decade

TAM · QUANTUM COMPUTING

\$16.3B by 2030

From \$2.2B (2025), 33.7% CAGR. Reaching \$19–50B by 2035 across market research estimates.

SAM · QEC / DECODING

\$400M+ raised

Capital raised across 8 QEC-focused vendors in the last 18 months alone (Riverlane, Q-CTRL, Quantum Machines, Algorithmiq).

GOVERNMENT BACKING

\$13B+ committed

US Dept. of Commerce \$2.0B (9 firms) + China \$10B national lab + EU Quantum Flagship €1B — QEC is on every roadmap.

Global quantum computing market, USD billions



Sources: market.us, Precedence Research, Grand View Research (TAM); TheQuantumInsider (QEC funding); NIST/Dept. of Commerce, PatentPC, EU Quantum Flagship (government).

Who pays, how we ship, how we make money

CUSTOMERS

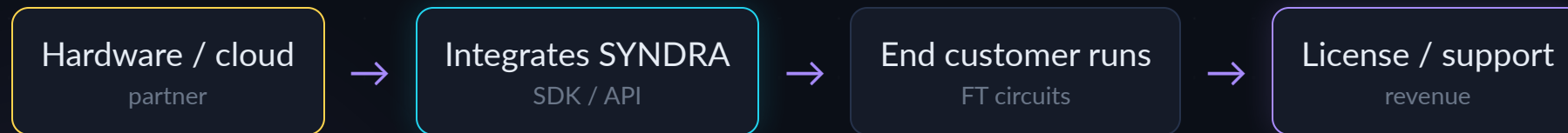
- ▶ Hardware makers (superconducting, photonic, trapped-ion) needing a real-time decoder.
- ▶ Cloud quantum-computing platforms (Braket/Azure Quantum-style offerings) serving enterprise users.
- ▶ National labs & research consortia validating FT roadmaps.

HOW WE SHIP

- ▶ **Decoder-as-a-service** SDK/API, integrated into a partner's control stack.
- ▶ **Open-source core** to drive adoption & community trust, matching Quandela/Riverlane's playbook.
- ▶ Direct co-development engagements with hardware OEMs.

HOW WE MONETIZE

- ▶ Per-decode / compute-time licensing for production deployments.
- ▶ Annual support & hardware-specific customization contracts.
- ▶ Grant & co-development funding (DARPA-style, national programs).



ILLUSTRATIVE PRICING Tiered licensing by decode volume, plus fixed annual support contracts — the QEC-vendor playbook (Riverlane, Q-CTRL): low fixed cost, scales with usage.

Where we sit against classical & proprietary decoders

DECODER	LATENCY	ADAPTS TO NOISE	SCALES (PARAMS)	OPENNESS
MWPM (classical)	medium	no	limited	mature, closed tooling
Google AlphaQubit	~63μs	yes	millions (NN)	proprietary
Riverlane Deltaflow	6.5μs (10× faster)	yes	large	proprietary, \$75M+ raised
SYNDRA (ours)	target: real-time	yes, native	O(log N), ~42–4k	open-source

OUR WEDGE Proprietary decoders (Google, Riverlane) are fast but closed and hardware-locked. We're **open-source**, **hardware-agnostic**, and **quantum-native** — the lane the incumbents leave open for the wider ecosystem.

O(log N)
PARAMS VS MILLIONS (NN)

0
HARDWARE LOCK-IN

100%
OPEN-SOURCE CORE

Sources: Google (Nature 2024, below-threshold QEC); Riverlane Deltaflow benchmark & funding (TheQuantumInsider, 2026); this project's own measured benchmark (Results).

From first results to industry-ready decoders

2026–2027 · RESEARCH & HARDWARE VALIDATION

2027–2028+ · SCALE & MARKET READINESS

1

2

3

4

2026 H2

Foundations

- ▶ Reproduce prior research & build the QCNN benchmark pipeline.
- ▶ First simulated results: QCNN vs classical (MWPM) decoding.
- ▶ Establish the metrics that matter (p_L , threshold, Λ).

2027 H1

Hardware Validation

- ▶ Run QCNN / VQE prototypes on real quantum hardware, e.g. via Quandela Cloud.
- ▶ Test across noise characteristics (superconducting, photonic).
- ▶ First pilot partnership signed with a hardware provider.

2027 H2 – 2028 H1

Scaling & Integration

- ▶ Add QGM; unify QCNN + VQE + QGM into one framework.
- ▶ Agentic system auto-selects method & parameters per problem.
- ▶ Seed/Series-A raise to fund scaling & hiring.

2028 H1+

Market & Industry Readiness

- ▶ Production-grade open-source release; first paying customers.
- ▶ Partner with hardware providers for deployment & validation.
- ▶ Position as the default decoder for FT quantum computing stacks.

Why fund SYNDRA

IMPACT

Faster, adaptive decoders shorten the path to useful fault-tolerant quantum computing — the bottleneck standing between today's noisy devices and applications in **cryptography, materials, and drug discovery**. Open-source lowers the barrier for the whole ecosystem, including emerging quantum hubs in Southeast Asia.

TRACTION (BUILT)

Working end-to-end pipeline: surface-code data → 4 decoders → paper-exact metrics, on live GPU infrastructure. Measured a transparent month-1 baseline against MWPM and a published Google reference — with a sourced plan to close the gap, and a reproducible, open-source codebase backing every number.

THE ASK

Compute credits, mentorship, and seed funding to execute the 2026–2028 roadmap: hardware validation on real quantum devices, scaling to larger code distances, and landing our first paying pilot customer among hardware & cloud QC providers.

1.34

MWPM Δ (MEASURED)

0.68

OUR QCNN Δ (MEASURED, M1)

→ 2.2

OUR TARGET Δ (PROJECTED)

\$16.3B

MARKET BY 2030

LET'S TALK We're raising to hit our first hardware-validated pilot in 2027. If you fund deep-tech quantum infrastructure, we want to talk roadmap and terms.

THANK YOU

SYNDRA

Quantum-Learning Decoders for Stabilizer Codes

TEAM

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TEAM

Nguyen Gia Hien

TEAM

**Hoang Dinh Duy
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TEAM

**Truong Le Gia
Khanh**

Open-source · hardware-agnostic · built for the next generation of fault-tolerant quantum computers. Let's talk roadmap, compute, and partnerships.

